

Microsoft Windows [Version 10.0.26200.9168]
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```
C:\Windows\System32>mongosh "mongodb+srv://cluster0.ruaqspr.mongodb.net/" --apiVersion 1
--username athineniaruntej_db_user
Enter password: *****
Current Mongosh Log ID: 6a8939c251ce9504b9d743cd
Connecting to:
mongodb+srv://<credentials>@cluster0.ruaqspr.mongodb.net/?appName=mongosh+2.10.0
Using MongoDB:      8.0.29 (API Version 1)
Using Mongosh:      2.10.0
```

For mongosh info see: <https://www.mongodb.com/docs/mongodb-shell/>

To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (<https://www.mongodb.com/legal/privacy-policy>).
You can opt-out by running the `disableTelemetry()` command.

```
Atlas atlas-3xqa9o-shard-0 [primary] test> use bookflow db.createCollection("book_metadata") show
collections
switched to db bookflow
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> use bookflow
already on db bookflow
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.createCollection("book_metadata")
{ ok: 1 }
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> show collections
book_metadata
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.book_metadata.insertOne({ title: "The Silent
Algorithm", author: "R. Menon", genre: "Technology", publishedYear: 2022, formats: ["Hardcover",
"Digital", "Audiobook"], specs: { pages: 340, fileSize_MB: 4.2, drm: true, narrator: "K. Rao",
runtime_min: 610 }, avgRating: 4.6, reviews: [ { member_id: "M1001", rating: 5, comment:
"Excellent deep dive into digital systems design.", date: ISODate("2023-01-15") }, { member_id:
"M1042", rating: 4, comment: "Good but the Digital edition had formatting issues.", date:
ISODate("2023-02-20") } ] });
{
  acknowledged: true,
  insertedId: ObjectId('6a893a8651ce9504b9d743ce')
}
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.book_metadata.insertMany([ { title: "Ocean of
Code", author: "A. Fernandes", genre: "Technology", publishedYear: 2021, formats: ["Digital",
"Paperback"], specs: { pages: 210, fileSize_MB: 2.1, drm: false }, avgRating: 4.2, reviews: [ {
member_id: "M2001", rating: 4, comment: "Solid Digital read, clean layout.", date: ISODate("2022-05-10")
}, { member_id: "M2002", rating: 5, comment: "Best programming book this year.", date:
ISODate("2022-06-01") } ] }, { title: "Whispers of Yesterday", author: "S. Kapoor", genre: "Fiction",
publishedYear: 2019, formats: ["Hardcover", "Paperback"], specs: { pages: 410, binding: "Sewn" },
avgRating: 3.8, reviews: [ { member_id: "M3001", rating: 3, comment: "Slow start but strong ending.",
date: ISODate("2020-01-15") }, { member_id: "M3002", rating: 4, comment: "Beautifully written prose.",
date: ISODate("2020-02-11") } ] }, { title: "Quantum Horizons", author: "R. Menon", genre:
"Science", publishedYear: 2023, formats: ["Digital", "Audiobook"], specs: { fileSize_MB: 5.6, drm:
true, narrator: "P. Iyer", runtime_min: 540 }, avgRating: 4.9, reviews: [ { member_id: "M4001",
rating: 5, comment: "The Digital format made annotations so easy.", date: ISODate("2023-08-01") }, {
member_id: "M4002", rating: 5, comment: "Mind-bending and well narrated.", date:
ISODate("2023-09-14") } ] } ] );
{
```

```
acknowledged: true,
insertedIds: {
  '0': ObjectId('6a893aa651ce9504b9d743cf'),
  '1': ObjectId('6a893aa651ce9504b9d743d0'),
  '2': ObjectId('6a893aa651ce9504b9d743d1')
}
}
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.book_metadata.countDocuments()
4
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.book_metadata.find().pretty()
[
  {
    _id: ObjectId('6a893a8651ce9504b9d743ce'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a893aa651ce9504b9d743cf'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
```

```
    date: ISODate('2022-05-10T00:00:00.000Z')
  },
  {
    member_id: 'M2002',
    rating: 5,
    comment: 'Best programming book this year.',
    date: ISODate('2022-06-01T00:00:00.000Z')
  }
]
},
{
  _id: ObjectId('6a893aa651ce9504b9d743d0'),
  title: 'Whispers of Yesterday',
  author: 'S. Kapoor',
  genre: 'Fiction',
  publishedYear: 2019,
  formats: [ 'Hardcover', 'Paperback' ],
  specs: { pages: 410, binding: 'Sewn' },
  avgRating: 3.8,
  reviews: [
    {
      member_id: 'M3001',
      rating: 3,
      comment: 'Slow start but strong ending.',
      date: ISODate('2020-01-15T00:00:00.000Z')
    },
    {
      member_id: 'M3002',
      rating: 4,
      comment: 'Beautifully written prose.',
      date: ISODate('2020-02-11T00:00:00.000Z')
    }
  ]
},
{
  _id: ObjectId('6a893aa651ce9504b9d743d1'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [ 'Digital', 'Audiobook' ],
  specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
  avgRating: 4.9,
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: ISODate('2023-08-01T00:00:00.000Z')
    },
    {
      member_id: 'M4002',
      rating: 5,
      comment: 'Mind-bending and well narrated.',
      date: ISODate('2023-09-14T00:00:00.000Z')
```

```

    }
  ]
}
]
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.book_metadata.find( { avgRating: { $gt: 4 } }, { title: 1,
avgRating: 1, _id: 0 } );
[
  { title: 'The Silent Algorithm', avgRating: 4.6 },
  { title: 'Ocean of Code', avgRating: 4.2 },
  { title: 'Quantum Horizons', avgRating: 4.9 }
]
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.book_metadata.find( { formats: { $in: ["Digital",
"Audiobook"] } }, { title: 1, formats: 1, _id: 0 } );
[
  {
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  { title: 'Ocean of Code', formats: [ 'Digital', 'Paperback' ] },
  { title: 'Quantum Horizons', formats: [ 'Digital', 'Audiobook' ] }
]
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.book_metadata.find({ avgRating: { $gt: 4 }, formats: {
$in: ["Digital"] } });
[
  {
    _id: ObjectId('6a893a8651ce9504b9d743ce'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {

```

```

    _id: ObjectId('6a893aa651ce9504b9d743cf'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      },
      {
        member_id: 'M2002',
        rating: 5,
        comment: 'Best programming book this year.',
        date: ISODate('2022-06-01T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a893aa651ce9504b9d743d1'),
    title: 'Quantum Horizons',
    author: 'R. Menon',
    genre: 'Science',
    publishedYear: 2023,
    formats: [ 'Digital', 'Audiobook' ],
    specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
    avgRating: 4.9,
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: ISODate('2023-08-01T00:00:00.000Z')
      },
      {
        member_id: 'M4002',
        rating: 5,
        comment: 'Mind-bending and well narrated.',
        date: ISODate('2023-09-14T00:00:00.000Z')
      }
    ]
  }
]
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.book_metadata.find( { formats: { $regex: "Digital",
$options: "i" } }, { title: 1, formats: 1 } );
[
  {
    _id: ObjectId('6a893a8651ce9504b9d743ce'),
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  }
]

```

```

},
{
  _id: ObjectId('6a893aa651ce9504b9d743cf'),
  title: 'Ocean of Code',
  formats: [ 'Digital', 'Paperback' ]
},
{
  _id: ObjectId('6a893aa651ce9504b9d743d1'),
  title: 'Quantum Horizons',
  formats: [ 'Digital', 'Audiobook' ]
}
]
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.book_metadata.find( { "reviews.comment": { $regex:
"Digital", $options: "i" } }, { title: 1, "reviews.$": 1 } );
[
  {
    _id: ObjectId('6a893a8651ce9504b9d743ce'),
    title: 'The Silent Algorithm',
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a893aa651ce9504b9d743cf'),
    title: 'Ocean of Code',
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a893aa651ce9504b9d743d1'),
    title: 'Quantum Horizons',
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: ISODate('2023-08-01T00:00:00.000Z')
      }
    ]
  }
]
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.book_metadata.aggregate([ { $match: {
publishedYear: { $gt: 2020 } } }, { $unwind: "$reviews" }, { $group: { _id: "$title",
avgReviewRating: { $avg: "$reviews.rating" }, totalReviews: { $sum: 1 }, author: { $first: "$author"

```

```
} } }], { $sort: { avgReviewRating: -1 } }, { $project: { _id: 0, title: "$_id", author: 1, avgReviewRating: { $round: ["$avgReviewRating", 2] }, totalReviews: 1 } } ]]);
```

```
[  
  {  
    totalReviews: 2,  
    author: 'R. Menon',  
    title: 'Quantum Horizons',  
    avgReviewRating: 5  
  },  
  {  
    totalReviews: 2,  
    author: 'R. Menon',  
    title: 'The Silent Algorithm',  
    avgReviewRating: 4.5  
  },  
  {  
    totalReviews: 2,  
    author: 'A. Fernandes',  
    title: 'Ocean of Code',  
    avgReviewRating: 4.5  
  }  
]
```

```
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.book_metadata.createIndex({ "reviews.comment":  
"text" });
```

```
reviews.comment_text
```

```
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.book_metadata.find({ $text: { $search: "Digital" } });
```

```
[  
  {  
    _id: ObjectId('6a893a8651ce9504b9d743ce'),  
    title: 'The Silent Algorithm',  
    author: 'R. Menon',  
    genre: 'Technology',  
    publishedYear: 2022,  
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],  
    specs: {  
      pages: 340,  
      fileSize_MB: 4.2,  
      drm: true,  
      narrator: 'K. Rao',  
      runtime_min: 610  
    },  
    avgRating: 4.6,  
    reviews: [  
      {  
        member_id: 'M1001',  
        rating: 5,  
        comment: 'Excellent deep dive into digital systems design.',  
        date: ISODate('2023-01-15T00:00:00.000Z')  
      },  
      {  
        member_id: 'M1042',  
        rating: 4,  
        comment: 'Good but the Digital edition had formatting issues.',  
        date: ISODate('2023-02-20T00:00:00.000Z')  
      }  
    ]  
  }  
]
```

```

    ]
  },
  {
    _id: ObjectId('6a893aa651ce9504b9d743d1'),
    title: 'Quantum Horizons',
    author: 'R. Menon',
    genre: 'Science',
    publishedYear: 2023,
    formats: [ 'Digital', 'Audiobook' ],
    specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
    avgRating: 4.9,
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: ISODate('2023-08-01T00:00:00.000Z')
      },
      {
        member_id: 'M4002',
        rating: 5,
        comment: 'Mind-bending and well narrated.',
        date: ISODate('2023-09-14T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a893aa651ce9504b9d743cf'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      },
      {
        member_id: 'M2002',
        rating: 5,
        comment: 'Best programming book this year.',
        date: ISODate('2022-06-01T00:00:00.000Z')
      }
    ]
  }
]
}
]
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.book_metadata.find( { $text: { $search: "Digital" } }, {
title: 1, score: { $meta: "textScore" } }).sort({ score: { $meta: "textScore" } });
[
  {

```



```

    _id: ObjectId('6a893a8651ce9504b9d743ce'),
    title: 'The Silent Algorithm',
    score: 1.1833333333333333
  },
  {
    _id: ObjectId('6a893aa651ce9504b9d743cf'),
    title: 'Ocean of Code',
    score: 0.6
  },
  {
    _id: ObjectId('6a893aa651ce9504b9d743d1'),
    title: 'Quantum Horizons',
    score: 0.6
  }
]
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.book_metadata.find( { $text: { $search: "Digital" } }
).explain("executionStats");
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookflow.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    },
    indexFilterSet: false,
    queryHash: 'CF6F4CEE',
    planCacheShapeHash: 'CF6F4CEE',
    planCacheKey: '08852285',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'TEXT_MATCH',
      indexPrefix: {},
      indexName: 'reviews.comment_text',
      parsedTextQuery: {
        terms: [ 'digit' ],
        negatedTerms: [],
        phrases: [],
        negatedPhrases: []
      }
    },
    textIndexVersion: 3,
    inputStage: {
      stage: 'FETCH',
      inputStage: {
        stage: 'IXSCAN',
        keyPattern: { _fts: 'text', _ftsx: 1 },

```

```
    indexName: 'reviews.comment_text',
    isMultiKey: true,
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'backward',
    indexBounds: {}
  }
},
rejectedPlans: [],
},
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 0,
  totalKeysExamined: 3,
  totalDocsExamined: 3,
  executionStages: {
    isCached: false,
    stage: 'TEXT_MATCH',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    indexPrefix: {},
    indexName: 'reviews.comment_text',
    parsedTextQuery: {
      terms: [ 'digit' ],
      negatedTerms: [],
      phrases: [],
      negatedPhrases: []
    },
  },
  textIndexVersion: 3,
  docsRejected: 0,
  inputStage: {
    stage: 'FETCH',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    docsExamined: 3,
    alreadyHasObj: 0,
    inputStage: {
```

```
stage: 'IXSCAN',
nReturned: 3,
executionTimeMillisEstimate: 0,
works: 4,
advanced: 3,
needTime: 0,
needYield: 0,
saveState: 0,
restoreState: 0,
isEOF: 1,
keyPattern: { _fts: 'text', _ftsx: 1 },
indexName: 'reviews.comment_text',
isMultiKey: true,
isUnique: false,
isSparse: false,
isPartial: false,
indexVersion: 2,
direction: 'backward',
indexBounds: {},
keysExamined: 3,
seeks: 1,
dupsTested: 3,
dupsDropped: 0
}
}
},
queryShapeHash:
'72B65B9E1BDB6FE0C468D2D50E4BCD0F9BF2A08FC1D5675B90B7C6DEFEBE2FAD',
command: {
  find: 'book_metadata',
  filter: { '$text': { '$search': 'Digital' } },
  '$db': 'bookflow'
},
serverInfo: {
  host: 'ac-skrervz-shard-00-01.ruaqspr.mongodb.net',
  port: 27017,
  version: '8.0.29',
  gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
  clusterTime: Timestamp({ t: 1787378850, i: 18 }),
```

```

signature: {
  hash: Binary.createFromBase64('eoANu/jTAkA2Szly9T/pAa3Wk98=', 0),
  keyId: Long('7642799070922145798')
},
operationTime: Timestamp({ t: 1787378850, i: 18 })
}
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.book_metadata.find( { "reviews.comment": { $regex:
"Digital", $options: "i" } } ).explain("executionStats");
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookflow.book_metadata',
    parsedQuery: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
    indexFilterSet: false,
    queryHash: '9DF7CB16',
    planCacheShapeHash: '9DF7CB16',
    planCacheKey: 'A9183C86',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
      direction: 'forward'
    },
    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 3,
    executionTimeMillis: 0,
    totalKeysExamined: 0,
    totalDocsExamined: 4,
    executionStages: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
      nReturned: 3,
      executionTimeMillisEstimate: 0,
      works: 5,
      advanced: 3,
      needTime: 1,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      direction: 'forward',
      docsExamined: 4
    }
  },
  queryShapeHash:

```

```
'7A8FA4C9B4F920C2612ACE0F25736A002251747FF102F1F0286A1CA539934508',
command: {
  find: 'book_metadata',
  filter: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
  '$db': 'bookflow'
},
serverInfo: {
  host: 'ac-skrervz-shard-00-01.ruagspr.mongodb.net',
  port: 27017,
  version: '8.0.29',
  gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
  clusterTime: Timestamp({ t: 1787379289, i: 5 }),
  signature: {
    hash: Binary.createFromBase64('vbN6D8FJXHt7gcrLy8jhlQPMjHs=', 0),
    keyId: Long('7642799070922145798')
  }
},
operationTime: Timestamp({ t: 1787379289, i: 5 })
}
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.book_metadata.createIndex({ formats: 1 });
db.book_metadata.createIndex({ avgRating: -1 });
avgRating_-1
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.book_metadata.find({ avgRating: { $gt: 4 }
}).explain("executionStats");
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookflow.book_metadata',
    parsedQuery: { avgRating: { '$gt': 4 } },
    indexFilterSet: false,
    queryHash: '85BA8276',
    planCacheShapeHash: '85BA8276',
    planCacheKey: '6EB4A3FD',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
```

```

stage: 'FETCH',
inputStage: {
  stage: 'IXSCAN',
  keyPattern: { avgRating: -1 },
  indexName: 'avgRating_-1',
  isMultiKey: false,
  multiKeyPaths: { avgRating: [] },
  isUnique: false,
  isSparse: false,
  isPartial: false,
  indexVersion: 2,
  direction: 'forward',
  indexBounds: { avgRating: [ '[inf.0, 4)' ] }
},
rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 1,
  totalKeysExamined: 3,
  totalDocsExamined: 3,
  executionStages: {
    isCached: false,
    stage: 'FETCH',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    docsExamined: 3,
    alreadyHasObj: 0,
    inputStage: {
      stage: 'IXSCAN',
      nReturned: 3,
      executionTimeMillisEstimate: 0,
      works: 4,
      advanced: 3,
      needTime: 0,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      keyPattern: { avgRating: -1 },
      indexName: 'avgRating_-1',
      isMultiKey: false,
      multiKeyPaths: { avgRating: [] },
      isUnique: false,
      isSparse: false,
      isPartial: false,

```

```

    indexVersion: 2,
    direction: 'forward',
    indexBounds: { avgRating: [ '[inf.0, 4)' ] },
    keysExamined: 3,
    seeks: 1,
    dupsTested: 0,
    dupsDropped: 0
  }
},
queryShapeHash:
'BD18AF34556307778EA6807595ED1C44E250FA13FE6A4A8DD51364D9D6A32DA9',
command: {
  find: 'book_metadata',
  filter: { avgRating: { '$gt': 4 } },
  '$db': 'bookflow'
},
serverInfo: {
  host: 'ac-skrervz-shard-00-01.ruaqspr.mongodb.net',
  port: 27017,
  version: '8.0.29',
  gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
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  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
  clusterTime: Timestamp({ t: 1787379349, i: 2 }),
  signature: {
    hash: Binary.createFromBase64('jsQhUgW7BUlxualjPs5MOfXpnAc=', 0),
    keyId: Long('7642799070922145798')
  }
},
operationTime: Timestamp({ t: 1787379349, i: 2 })
}
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.book_metadata.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  {
    v: 2,
    key: { _fts: 'text', _ftsx: 1 },
    name: 'reviews.comment_text',
    weights: { 'reviews.comment': 1 },
    default_language: 'english',
    language_override: 'language',

```

```
textIndexVersion: 3
},
{ v: 2, key: { formats: 1 }, name: 'formats_1' },
{ v: 2, key: { avgRating: -1 }, name: 'avgRating_-1' }
]
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> show dbs
bookflow 132.00 KiB
admin      0 B
local      0 B
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> show collections
book_metadata
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.book_metadata.countDocuments()
4
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> db.book_metadata.find( {}, { title: 1, avgRating: 1, _id: 0 }
);
[
  { title: 'The Silent Algorithm', avgRating: 4.6 },
  { title: 'Ocean of Code', avgRating: 4.2 },
  { title: 'Whispers of Yesterday', avgRating: 3.8 },
  { title: 'Quantum Horizons', avgRating: 4.9 }
]
Atlas atlas-3xqa9o-shard-0 [primary] bookflow> exit

C:\Windows\System32>
```